

ANALOG AND DIGITAL ELECTRONICS LAB

Experiment Theory

Experiment : MOD-10 Asynchronous Counter

A MOD-10 asynchronous counter, also called a decade counter, is a sequential digital circuit that counts through ten distinct states, from 0000 (decimal 0) to 1001 (decimal 9), and then automatically returns to 0000. It is generally constructed using four flip-flops connected in cascade, where the output of one flip-flop provides the clock input to the next. Since the clock signal is applied only to the first flip-flop and the remaining flip-flops are triggered by the outputs of preceding stages, it is called an asynchronous (ripple) counter. The counter operates in binary form for the first ten states.

To obtain MOD-10 operation, the counter is designed to reset when the count reaches 1010 (decimal 10). A suitable reset logic circuit detects this state and asynchronously clears all the flip-flops, causing the counter to return to 0000. Thus, the counter has 10 valid counting states and can be used in applications such as frequency division, digital clocks, event counting, and timing circuits. The counting sequence can be observed using LEDs or a logic analyser connected to the flip-flop outputs.

Experiment : Ring Counter and Johnson Counter

A ring counter is a type of shift-register-based counter in which the output of the last flip-flop is directly fed back to the input of the first flip-flop. It is generally constructed using a number of flip-flops connected in cascade and operates by circulating a single logic 1 through the flip-flops. For an n -bit ring counter, there are n valid states, provided the counter is initially preset with one flip-flop at logic 1 and the remaining flip-flops at logic 0. With each clock pulse, the logic 1 shifts to the next flip-flop, producing a sequence of mutually exclusive output pulses. Ring counters are commonly used for sequence generation, timing control, and step-by-step control applications.

A Johnson counter, also known as a twisted-ring counter, is a modified ring counter in which the complemented output of the last flip-flop is fed back to the input of the first flip-flop. An n -bit Johnson counter produces $2n$ distinct states, making it more efficient in terms of the number of states obtained from the same number of flip-flops. The counter passes through a predetermined sequence of states as clock pulses are applied and then repeats continuously. Both ring and Johnson counters are useful in digital systems for sequence generation, timing, frequency division, and control applications.